Università degli Studi di Cagliari
Faculty of Economic, Legal and Political Sciences
Master in Economics, Finance and Data Analysis

Spatial and Regional Economics (Module A)

Homework 1: The Rosen–Roback Spatial Equilibrium Model

Academic Year 2026–2027

Team Members:
Davide Maiellaro Caterina Congia
Federica Stella Favalli Lorenzo Piva

October 14, 2026

Instructor: Prof. Fabio Cerina | **Discussion & Presentation Date:** October 14, 2026

AI Disclosure Statement. The team used two AI assistants: Google Gemini (first draft of the LaTeX text and debugging of the Matlab block) and Claude by Anthropic (review of the derivations, consistency check of the reported numbers, LaTeX editing). All derivations, numerical results and interpretations were checked by the authors, who re-ran every script reported here.

Part A – The Puzzle: Model without Land (Topic 2)

Throughout, $n = 60$, $\bar{L} = 10^5$, $\gamma = 0.3$, $K_i = 1$, seed `rng(7)`. The model and its inversion on data (L_i, w_i) are

$$w_i = A_i K_i^{\gamma} L_i^{-\gamma}, \qquad U_i = w_i B_i = W \quad \Longrightarrow \quad \hat{A}_i = w_i L_i^{\gamma} K_i^{-\gamma}, \qquad \hat{B}_i = \frac{W}{w_i}, \tag{1}$$

with W fixed by the normalization $\frac{1}{n}\sum_i \hat{B}_i = 1$ (function `rr_invert`). In logs, $\ln \hat{B}_i = \ln W - \ln w_i$: recovered amenities are the mirror image of wages.

(a) The population–amenities panel in the original script

What the script does. The original data-generating block draws $L_i \sim \text{Uniform}(0, 1)$, rescaled so that $\sum_i L_i = \bar{L}$, and, *independently*, $w_i \sim \text{Uniform}(0.5, 2)$. It then inverts these data with (1) and plots the four panels of Figure 1. The printed OLS slopes are in Table 1.

Table 1: OLS slopes, original data-generating block

Panel	**Regression (logs)**	**Slope**
Population vs productivity	$\ln L_i$ on $\ln \hat{A}_i$	1.333
Population vs amenities	$\ln L_i$ on $\ln \hat{B}_i$	**0.192**
Wages vs productivity	$\ln w_i$ on $\ln \hat{A}_i$	0.600
Wages vs amenities	$\ln w_i$ on $\ln \hat{B}_i$	−1.000

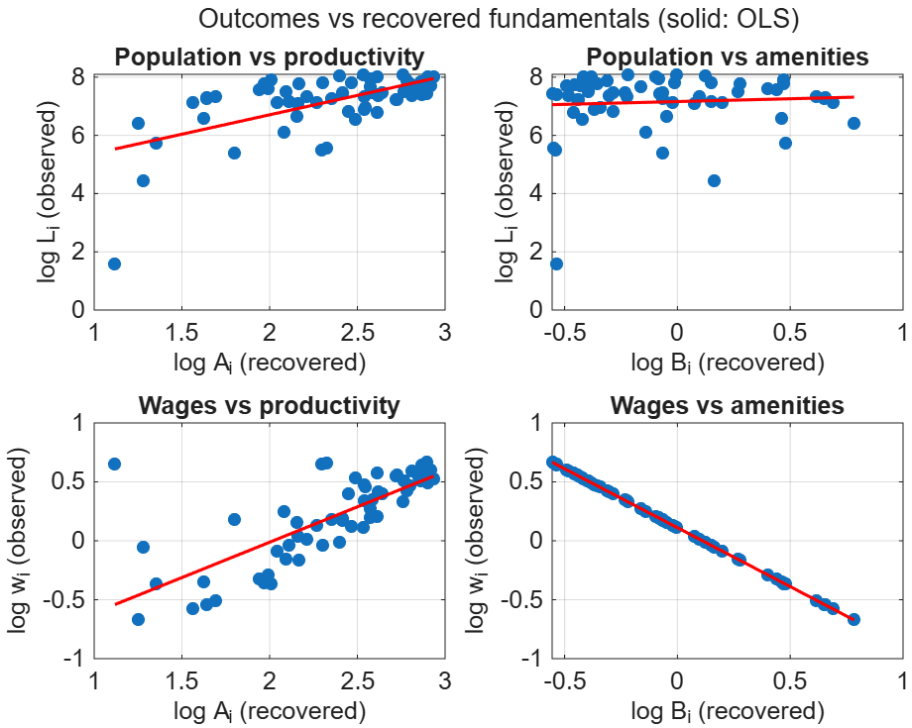


Figure 1: Original script (L and w drawn independently). Top right: population vs recovered amenities.

Description of the panel.

- **Sign:** slightly positive, $\hat{\beta}_{L,B} = 0.192$.
- **Size:** economically and statistically zero. A 1% higher recovered amenity is associated with only 0.19% more population. The correlation is $\text{corr}(\ln L, \ln \hat{B}) = +0.07$, so $R^2 \approx 0.005$ and the t-statistic is $t = r\sqrt{n-2}/\sqrt{1-r^2} \approx 0.5$. The points form a shapeless cloud, even though both variables vary a lot: $\ln L_i$ goes from about 1.6 to 8.1 and $\ln \hat{B}_i$ from -0.56 to 0.78, so amenities differ by a factor of almost 4.

Why this is surprising given $U_i = w_i B_i$. People value amenities, since $\partial U_i / \partial B_i = w_i > 0$. Mobility should therefore send them to high-B places. Indeed, in equilibrium (Topic 2, eq. 1) population shares are

$$\pi_i = \frac{L_i}{\bar{L}} = \frac{K_i (A_i B_i)^{1/\gamma}}{\sum_j K_j (A_j B_j)^{1/\gamma}}, \tag{2}$$

increasing in B_i. The same holds for the recovered fundamentals: multiplying the two formulas in (1) gives $\hat{A}_i \hat{B}_i = W L_i^{\gamma}$ (with $K_i = 1$), i.e.

$$\ln L_i = \tfrac{1}{\gamma}\left(\ln \hat{A}_i + \ln \hat{B}_i\right) - \tfrac{1}{\gamma} \ln W. \tag{3}$$

For two locations with the same $\hat{A}$, the one with 1% higher amenities must be $1/\gamma = 3.33\%$ larger. The evidence shown in Topic 2 points the same way: in Glaeser and Gottlieb (2009), metropolitan population growth rises with mean January temperature, a classic amenity. So we would expect a clearly upward-sloping panel. Instead we get a flat one: this is the anomaly.

Where the flat slope comes from. Since $\ln \hat{B}_i = \ln W - \ln w_i$, the OLS slope of the panel is (derived in part (d))

$$\hat{\beta}_{L,B} = -\frac{\widehat{\text{Cov}}(\ln L_i, \ln w_i)}{\widehat{\text{Var}}(\ln w_i)}. \tag{4}$$

The panel therefore only reflects how L and w co-move in the data. In the original block they are drawn independently, so $\mathrm{Cov}(\ln L, \ln w) = 0$ and the expected slope is zero. The value 0.192 is pure sampling noise: in this draw $\mathrm{corr}(\ln L, \ln w) = -0.07$. Note that the panel does not hold $\hat{A}$ fixed. Equation (3) holds for any data, but in these data a high $\hat{B}$ (a low wage) also means a low $\hat{A}$, and the two effects cancel.

(b) Removing the assumption: the urban wage premium

Which assumption, which fact.

- **Assumption:** population and wages are drawn independently, which imposes

$$\mathrm{Cov}(\ln L_i, \ln w_i) = 0.$$

  Big and small locations pay, on average, the same wage.

- **Fact:** Topic 1, Fact 4, the urban wage premium. Denser places pay more (Autor, Ely Lecture 2019): log hourly wages rise steadily with log population density across US commuting zones.

Choice of ϕ. We replace the independent draw with $\ln w_i = \phi \ln L_i + \text{noise}$. In our model all locations have the same capital ($K_i = 1$) and there is no land, so size and density coincide up to a constant, and the elasticity with respect to density can be used as ϕ. A back-of-the-envelope reading of the fitted lines in Autor's figure (Topic 1, slide 22) gives the following. Over about three orders of magnitude of density (≈ 7 log points), log wages of college workers rise by about 0.5–0.6, an elasticity of about 0.07–0.08. For non-college workers the gradient was similar in 1970 and has since fallen to about 0.02. We set

$$\phi = 0.08, \qquad \sigma_\varepsilon = 0.15.$$

The noise term captures wage differences unrelated to size (industry mix, skills). With $\sigma_\varepsilon = 0.15$ the drawn data have $\mathrm{corr}(\ln L, \ln w) \approx 0.5$: a clear but noisy premium, as in the data. As shown in (d), σ_ε only affects the *size* of the slope, never its sign. Any $\phi > 0$ gives the same qualitative answer.

Modified block.

```
ORIGINAL_DGP = false;
phi   = 0.08;            % elasticity of wages to city size (Fact 4, Topic 1)
sig_e = 0.15;            % sd of the wage noise (set to 0 for Part A(d))
L_obs = rand(1, n);                       % same L draw in every case
L_obs = Ltot * L_obs / sum(L_obs);        % rescale so that sum(L) = Ltot
if ORIGINAL_DGP
    w_obs = 0.5 + 1.5 * rand(1, n);       % original: w uniform on [0.5, 2]
else
    lL    = log(L_obs);
    w_obs = exp(phi * (lL - mean(lL)) + sig_e * randn(1, n));
end
```

In formulas: $\ln w_i = \phi\,(\ln L_i - \overline{\ln L}) + \sigma_\varepsilon \varepsilon_i$, with $\varepsilon_i \sim \mathcal{N}(0,1)$ independent of L_i. Three remarks on the code:

- The switch `ORIGINAL_DGP = true` reproduces part (a) exactly, so the same file generates every result in this report.

- The seed and the L draw come first and are unchanged, so L_i is *identical* to part (a). Only wages change, and the comparison is clean (same vertical axis in Figures 1 and 2).
- Subtracting $\overline{\ln L}$ only sets the level of wages (geometric mean around 1, as in the original range $[0.5, 2]$). A constant drops out of every slope.

Prediction before running. Substituting the new wage equation into the inversion formula $\ln \hat{B}_i = \ln W - \ln w_i$:

$$\ln \hat{B}_i = \text{const} - \phi \ln L_i - \sigma_\varepsilon \varepsilon_i \qquad \Longrightarrow \qquad \frac{\partial \ln \hat{B}_i}{\partial \ln L_i} = -\phi = -0.08 < 0. \tag{5}$$

The panel puts $\ln L$ on the vertical axis and $\ln \hat{B}$ on the horizontal one. The sign of an OLS slope is the sign of the covariance, whichever variable is on the vertical axis. Since $\mathrm{Cov}(\ln L, \ln \hat{B}) = -\phi \,\mathrm{Var}(\ln L) < 0$, **we predict a negative slope**.

We can also predict its size. Using the population moments,

$$\beta_{L,B} = \frac{-\phi \, v_L}{\phi^2 v_L + \sigma_\varepsilon^2}, \qquad v_L \equiv \mathrm{Var}(\ln L_i). \tag{6}$$

With L_i uniform, $-\ln L_i$ is (up to a constant) exponentially distributed with variance 1, so $v_L \approx 1$ and $\beta_{L,B} \approx -0.08/(0.0064 + 0.0225) \approx -2.8$.

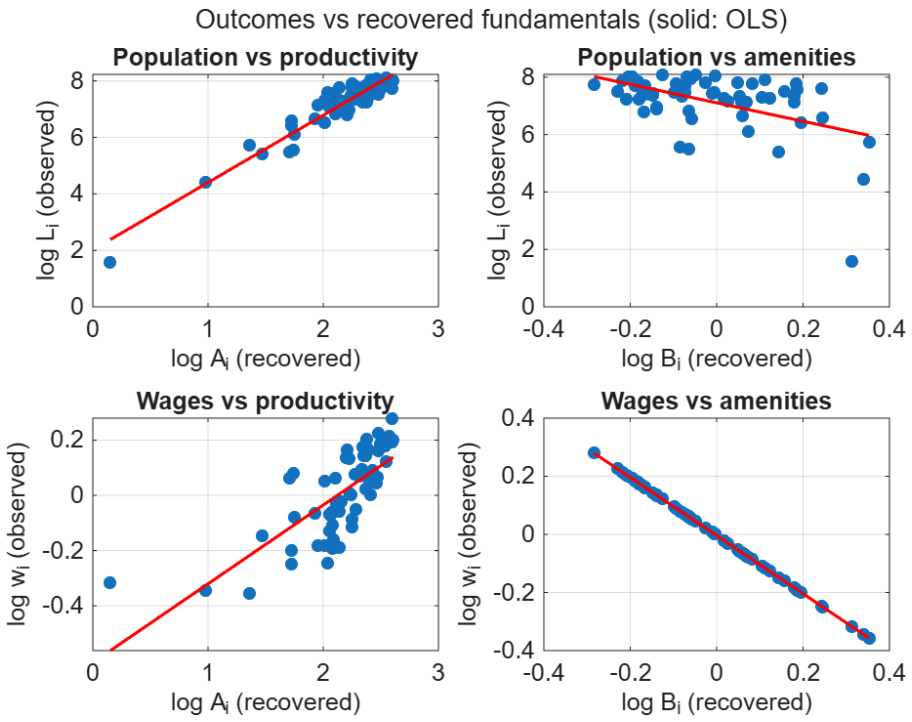


Figure 2: Modified block: $\ln w_i = \phi \ln L_i + \text{noise}$, $\phi = 0.08$, $\sigma_\varepsilon = 0.15$. Top right: the population–amenities panel is now downward sloping.

Results. The prediction is confirmed (Figure 2, Table 2). The population–amenities slope turns clearly negative, $\hat{\beta}_{L,B} = -3.217$, close to the predicted -2.8. The gap is sampling variability with $n = 60$, discussed in (d). The wages–amenities panel is still exactly -1, because $\ln \hat{B}_i = \text{const} - \ln w_i$ by construction. The population–productivity panel becomes steeper and tighter, because $\ln \hat{A}_i = \ln w_i + \gamma \ln L_i$ now increases with L_i through both terms. The wages–productivity slope falls toward $\phi/(\phi + \gamma) \approx 0.21$, because most of the variation in $\hat{A}$ now comes from L rather than from wage noise (see (d)).

Table 2: OLS slopes: original vs modified data-generating block

Panel	**Original (a)**	**Modified (b)**
Population vs productivity	1.333	2.382
Population vs amenities	**+0.192**	**−3.217**
Wages vs productivity	0.600	0.285
Wages vs amenities	−1.000	−1.000
$\text{corr}(\ln L, \ln w)$	−0.07	+0.49

What we conclude about the model without land as a measurement device.

1. **The inversion always fits, so it cannot flag the problem.** Any (L, w) is rationalized exactly (Topic 2: "inversion measures, it does not test"). The output is only as good as the model behind it.

2. **With a single price, amenities are just the inverse of wages.** Since $\hat{B}_i \propto 1/w_i$, anything that raises wages in large places, such as the urban wage premium, is read as compensation for bad amenities. Once the data respect Fact 4, the model mechanically concludes that the largest cities are the least pleasant places to live.

3. **The missing price is rent.** Topic 1 (Fact 4, cont.) also shows that denser places are more expensive: the elasticity of rents to density is about 0.16. In reality, high wages in big cities largely compensate for high housing costs, not for bad amenities. A model in which the wage is the only price cannot tell an *expensive* place from an *unpleasant* one.

Hence the model without land gives biased amenity measures, which are unlikely to pass an external validation against observable amenities such as climate or coastline (Topic 2, slide 24). Housing must be added, so that amenities are read off rents *and* wages, $B_i \propto r_i^{1-\mu}/w_i$ (Topic 3, Part B).

(c) Amenity spillovers: total vs exogenous amenities

Suppose $B_i = b_i L_i^{\eta}$, with b_i exogenous (climate, coast) and η the spillover elasticity. If $\eta > 0$, size creates amenities (restaurants, culture, leisure). If $\eta < 0$, size creates disamenities (congestion, pollution) (Topic 1, slide 32).

Panel with recovered total amenities $\hat{B}_i$: it does not change. The mobility condition $w_i B_i = W$ involves only *total* amenities: workers do not care whether B_i comes from nature or from other people. The inversion $\hat{B}_i = W/w_i$ uses only observed wages, and η appears nowhere. Given the same data (L_i, w_i), $\hat{B}_i$ is numerically identical and the panel is unchanged:

$$\frac{\partial \ln \hat{B}_i}{\partial \ln L_i} = -\phi \qquad \text{(same as in (b))}. \tag{7}$$

Panel with recovered exogenous amenities $\hat{b}_i = \hat{B}_i L_i^{-\eta}$: it changes. Taking logs and using (5),

$$\ln \hat{b}_i = \ln \hat{B}_i - \eta \ln L_i = \text{const} - (\phi + \eta) \ln L_i - \sigma_\varepsilon \varepsilon_i \qquad \Longrightarrow \qquad \frac{\partial \ln \hat{b}_i}{\partial \ln L_i} = -(\phi + \eta). \tag{8}$$

Each point moves horizontally by $-\eta \ln L_i$, more for larger locations. The sign of the panel is the sign of $-(\phi + \eta)$:

- **Positive spillovers ($\eta > 0$):** $-(\phi+\eta) < -\phi < 0$. Exogenous amenities fall even faster with size. Big cities already get a low total $\hat{B}$. If their size itself adds amenities, their natural component must be even lower to deliver the same low total. The panel stays negative, with a stronger negative correlation.
- **Congestion ($\eta < 0$):** the relation becomes less negative. It turns positive only if $\eta < -\phi = -0.08$, i.e. only if the congestion elasticity is larger in absolute value than the urban wage premium.

Note that the derivative (8) pins down the *sign* of the panel, not the slope of the fitted line. With $\ln L$ on the vertical axis and no noise, the fitted slope would be $-1/(\phi+\eta)$, as in (d).

Conclusion. Spillovers do not solve the puzzle. The panel with total amenities is unchanged, and the panel with exogenous amenities becomes positive only under strong congestion. Moreover, η cannot be learned from (L_i, w_i): any value of η rationalizes the same data by adjusting $\hat{b}_i$. Separating b_i from L_i^{η} requires information beyond the cross-section (Part C(c)).

(d) Bonus: the slope formula, zero noise, and attenuation

1. General formula. The OLS slope of $\ln L_i$ on $\ln \hat{B}_i$ is $\widehat{\mathrm{Cov}}(\ln \hat{B}, \ln L)/\widehat{\mathrm{Var}}(\ln \hat{B})$. Since $\ln \hat{B}_i = \ln W - \ln w_i$ and $\ln W$ is the same constant for all i,

$$\hat{\beta}_{L,B} = \frac{\widehat{\mathrm{Cov}}(\ln W - \ln w_i,\, \ln L_i)}{\widehat{\mathrm{Var}}(\ln W - \ln w_i)} = -\frac{\widehat{\mathrm{Cov}}(\ln L_i, \ln w_i)}{\widehat{\mathrm{Var}}(\ln w_i)}. \tag{9}$$

This is an exact identity in the sample. The script now prints $-\widehat{\mathrm{Cov}}/\widehat{\mathrm{Var}}$ next to the slope, and the two coincide: 0.192 in (a) and −3.217 in (b).

2. Zero noise: the slope is exactly $-1/\phi$. With $\sigma_\varepsilon = 0$, $\ln w_i = \phi(\ln L_i - \overline{\ln L})$, so

$$\ln \hat{B}_i = \text{const} - \phi \ln L_i \quad \Longleftrightarrow \quad \ln L_i = \text{const}' - \tfrac{1}{\phi} \ln \hat{B}_i. \tag{10}$$

All points lie exactly on a line with slope $-1/\phi$, so OLS returns it exactly. Equivalently, $\widehat{\mathrm{Cov}}(\ln L, \ln w) = \phi\, \hat{v}_L$ and $\widehat{\mathrm{Var}}(\ln w) = \phi^2 \hat{v}_L$, so $\hat{\beta}_{L,B} = -1/\phi = -12.5$, whatever the draw of L. **Verification** (script with `sig_e = 0`):

Panel	Slope	Theory
Population vs productivity	2.632	$1/(\phi+\gamma) = 1/0.38$
Population vs amenities	**−12.500**	$-1/\phi$
Wages vs productivity	0.211	$\phi/(\phi+\gamma) = 0.08/0.38$
Wages vs amenities	−1.000	by construction

The other two panels also match theory. Without noise, $\ln \hat{A}_i = \ln w_i + \gamma \ln L_i = (\phi+\gamma)\ln L_i + \text{const}$, which gives slopes $1/(\phi+\gamma)$ and $\phi/(\phi+\gamma)$.

3. Noise back on: the slope moves toward zero. With $\sigma_\varepsilon = 0.15$ the slope goes from −12.5 to −3.217: still negative, but much flatter. The reason is attenuation. The noise enters the *horizontal* variable, $\ln \hat{B}_i = \text{const} - \phi \ln L_i - \sigma_\varepsilon \varepsilon_i$, but is unrelated to $\ln L_i$. It spreads the points horizontally without moving them vertically. In terms of moments, it leaves the covariance unchanged and inflates the variance of the regressor:

$$\mathrm{Cov}(\ln L, \ln \hat{B}) = -\phi\, v_L, \qquad \mathrm{Var}(\ln \hat{B}) = \phi^2 v_L + \sigma_\varepsilon^2 \quad \Longrightarrow \quad \beta_{L,B} = -\frac{1}{\phi} \cdot \underbrace{\frac{\phi^2 v_L}{\phi^2 v_L + \sigma_\varepsilon^2}}_{\lambda \in (0,1)}. \tag{11}$$

With the sample $\hat{v}_L = 1.09$, the attenuation factor is $\lambda \approx 0.24$ and $\beta_{L,B} \approx -2.95$. The realized -3.217 differs because, with $n = 60$, the drawn noise is not exactly uncorrelated with $\ln L$ and does not have exactly variance σ_ε^2. The sample identity of point 1 holds exactly anyway. As σ_ε grows (or $\phi \to 0$), $\lambda \to 0$ and the slope goes to zero, which is the situation of part (a).

Part B – The Resolution: Model with Land (Topic 3)

[To be completed with `RR2_hw_amenities_panel.m`*.]*

Part C – Spillovers

[To be completed.]